

Chess Rating Estimation — Progress Notes

October 7, 2026

1 Setting

Classify the rating bin of a game from the game itself (moves, clocks, opening, time control). One row per game, label `elo_bin` from the average Elo of both players. Data: Lichess standard rated rapid games, April–June 2017. Model: XGBoost. Headline metric: accuracy; per-bin accuracy and the confusion matrix as diagnostics.

2 Sampling

34.6M games read, 7.64M rapid, 4.53M eligible. Filters: rated, rapid ($480 \leq \text{base} + 40 \cdot \text{inc} \leq 1499$), no bots, no abandoned or rules-infraction games, $|\Delta\text{Elo}| \leq 100$, at least 20 half-moves. Stratified reservoir sampling over 9 bins of 200 Elo, cap 50 000 per bin (333 478 games; 13 h single-threaded).

Bin	Kept	Eligible	Retention
< 800	47	47	100 %
[800, 1000)	22 340	22 340	100 %
[1000, 1200)	50 000	232 583	21.5 %
[1200, 1400)	50 000	762 977	6.5 %
[1400, 1600)	50 000	1 383 785	3.6 %
[1600, 1800)	50 000	1 329 333	3.8 %
[1800, 2000)	50 000	662 901	7.5 %
[2000, 2200)	50 000	121 233	41.2 %
≥ 2200	11 091	11 091	100 %

The tails are exhausted and stay below the target. For modelling the bins are merged to 7 classes: < 1000 (from < 800 and $[800, 1000)$) and ≥ 2000 (from $[2000, 2200)$ and ≥ 2200).

3 Train/test split

The label is mostly a property of the player, so train and test must be player-disjoint; otherwise the model can recognise players by their play. A game whose players land on different sides is dropped.

- 91 620 players; 31% have one game, but players with ≥ 10 games hold 68% of all player slots. 39% of players span more than one bin.
- Player graph (edge = played each other): 97.3% of players and 99.5% of games are in one connected component, so a split cannot follow components (Fig. 1).

- With independent assignment and fractions p_i , a game survives with probability $\sum_i p_i^2$ and lands in split i with probability p_i^2 . An 80/10/10 player split gives a 97/1.5/1.5 game split and loses 34%. Graph partitioning could lose less; not measured.

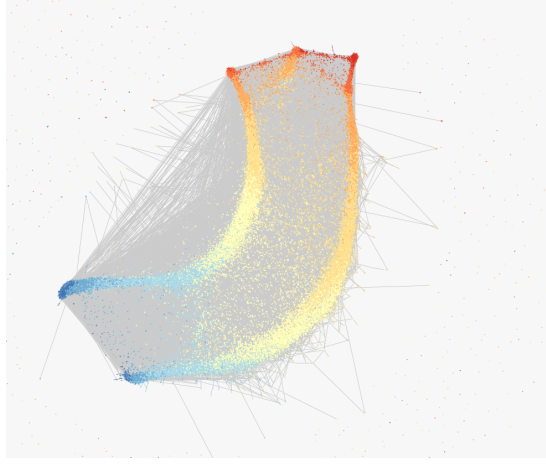


Figure 1: Player graph, nodes coloured by rating bin.

Test size. $SE \leq 0.5/\sqrt{n}$. About 10 000 test games give ± 1.0 pp at most (95%, independent). Games of one player are correlated: with $\bar{m} \approx 2.5$ test games per player and an assumed within-player error correlation $\rho = 0.1\text{--}0.5$, the interval widens by $\sqrt{1 + (\bar{m} - 1)\rho} \approx 1.07\text{--}1.32$, so roughly ± 1.3 pp. This is approximate, since a game has two players. Test fraction $p_{\text{test}}^2 \cdot 333\,478 \approx 10\,000$ gives $p_{\text{test}} \approx 0.173$.

Result. Independent assignment within 9 strata (quantiles of player mean Elo), seed 42, train/test = 0.827/0.173.

	Games	Players	Share of kept
Train	229 402	71 198	96.0%
Test	9 666	7 688	4.0%
Dropped	94 410 (28.3%)		

Retention 71.7% (predicted 71.4%). Disjointness check passes. Test ratings are 7–35 Elo lower than train across the deciles (mean 1563 vs. 1578; KS 0.023).

Games per class after merging:

	<1000	1000–1200	1200–1400	1400–1600	1600–1800	1800–2000	≥ 2000
Train	14 851	34 545	34 514	34 430	34 389	33 862	42 811
Test	778	1 449	1 421	1 417	1 416	1 576	1 609

Baselines (test). Majority class (≥ 2000): 16.6%. Uniform: 14.3%. Every model accuracy is reported next to these.

Validation. No fixed validation set (it would pay the p_i^2 cost again). Hyperparameters use 5-fold CV inside train with folds assigned by player. A game is a validation row of fold k if both players are in k , a fit row if neither is, and is used for fitting in the other folds if its players are in different folds. Each validation fold then holds only about $1/25$ of the games ($\approx 9\,000$).

Rejected.

- Temporal split: same players on both sides, leaks the label.

- Random 80/10/10 player split: see above.
- Split the eligible pool first, then sample: protects the tails, but means redoing the 13h sampling run.

4 Features

Excluded (label or identity): `white_elo`, `black_elo`, `avg_elo`, `rating_diff`, `white`, `black`, `game_id`, `utc_date`, `utc_time`, `split`. Current inputs: `result`, `termination`, `half-moves`, `moves`, `clocks`, `opening_family`, `tc_base`, `tc_increment`. Anything fitted on data (e.g. opening target encoding) is fitted on training rows only, inside each CV fold.

Stockfish (in progress). Engine features (centipawn loss, blunders, best-move agreement) for all train/test games, run on a rented machine with a hard cap of \$20. Calibration on 350 games (50 per bin), depths 15/18/20 and MultiPV 1–3: per-game ACPL vs. Elo has Spearman $\rho \approx -0.52$ for every configuration, with overlapping 95% CIs, so no measurable gain from more depth or MultiPV. Depth 15, MultiPV 1 is the leading candidate (est. \$5, unverified cloud price); final choice open.

5 Open

- Engine depth/MultiPV decision and the full remote run.
- Remaining PGN features (clock usage, development, castling).
- Class weights for the sparse < 1000 class.
- CV fold construction.
- Baseline XGBoost, then lift from engine features.